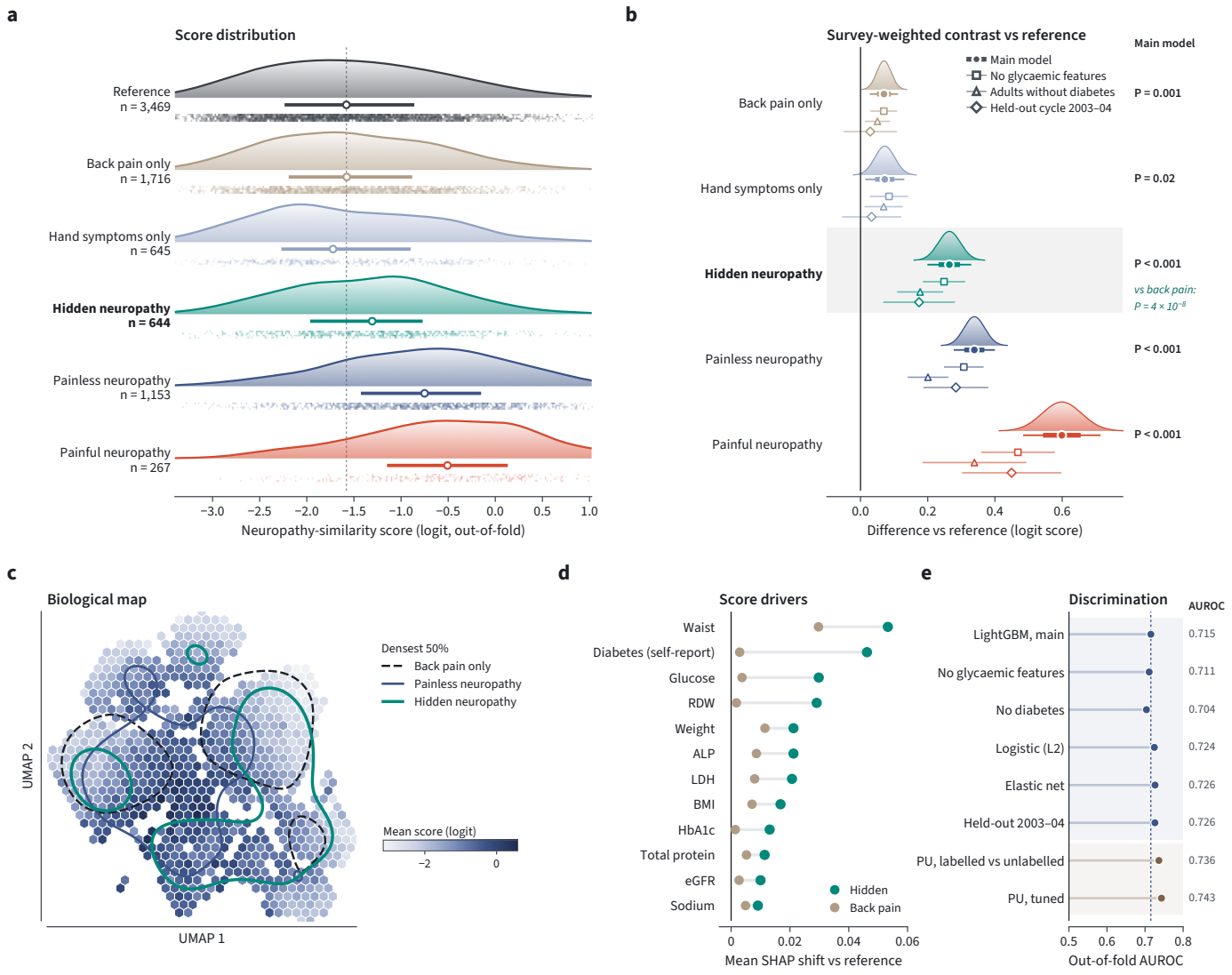

a
Score distribution
Reference
n = 3,469
Back pain only
n = 1,716
Hand symptoms only
n = 645
Hidden neuropathy
n = 644
Painless neuropathy
n = 1,153
Painful neuropathy
n = 267
−3.0 −2.5 −2.0 −1.5 −1.0 −0.5 0.0 0.5 1.0
Neuropathy-similarity score (logit, out-of-fold)
b
Survey-weighted contrast vs reference
Main model
No glycaemic features
Adults without diabetes
Held-out cycle 2003–04
Back pain only
Hand symptoms only
Hidden neuropathy
Painless neuropathy
Painful neuropathy
Main model
P = 0.001
P = 0.02
P < 0.001
vs back pain:
P = 4 × 10−8
P < 0.001
P < 0.001
0.0 0.2 0.4 0.6
Difference vs reference (logit score)
c
Biological map
Densest 50%
Back pain only
Painless neuropathy
Hidden neuropathy
Mean score (logit)
−2 0
UMAP 2
UMAP 1
d
Score drivers
Waist
Diabetes (self-report)
Glucose
RDW
Weight
ALP
LDH
BMI
HbA1c
Total protein
eGFR
Sodium
Hidden
Back pain
0 0.02 0.04 0.06
Mean SHAP shift vs reference
e
Discrimination
AUROC
LightGBM, main 0.715
No glycaemic features 0.711
No diabetes 0.704
Logistic (L2) 0.724
Elastic net 0.726
Held-out 2003–04 0.726
PU, labelled vs unlabelled 0.736
PU, tuned 0.743
0.5 0.6 0.7 0.8
Out-of-fold AUROC